Swayam Jaiswal

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Education

- **International Institute of Information Technology** — Pune, India
  Bachelor of Engineering - Computer Engineering; GPA: 9.19 — *Nov 2021 - May 2026*
  Courses: *Data Science and Big Data Analytics, Operating Systems, Data Structures and Algorithms, Artificial Intelligence, Computer Networks and Security, Database Management Systems, Discrete Mathematics*

Skills Summary

- **Languages**: Python, C++, Kotlin, Bash
- **Frameworks**: Langgraph, Scikit, NLTK, PyTorch, TensorFlow, Flask, FastAPI, HuggingFace, Langchain, FastMCP
- **AI/ML Skills**: Generative AI, Machine Learning, Deep Learning, NLP, RAG, Fine-Tuning, CNN, Prompt Engineering, Agentic AI, Model Context Protocol
- **Databases**: **SQL:** MySQL, OracleDB, SQLite **NoSQL**: MongoDB, **VectorDB :** FAISS, ChromaDB, Qdrant
- **Tools**: GIT, GitHub, Android Studio, Docker

Experience

- **VPN Digital Services** — Remote
  Android Developer Intern — *Feb 2024 - Apr 2024*
  - **Developed** the Android application prototype for *Rudrastra*, an OSINT tool, using Kotlin and Jetpack Compose, ensuring a responsive UI and seamless user experience. Integrated multiple API calls to reflect the tool's capabilities, collaborating with backend teams for API integration.
  - **Implemented** key features, including login and sign-up functionality, utilizing Firebase for secure authentication and storage, enhancing the app's usability and scalability for investor presentations.
  - **Delivered** a functional prototype, enabling the organization to showcase an Android app alongside their web application, successfully attracting potential investors and advancing the project's funding prospects.

Projects

- **Exam Helper AI (Full-Stack, Generative AI, LangChain, Together AI, ChromaDB, React, Flask, TypeScript)** ***(Individual)*** **Github** : Designed and implemented a full-stack web application that streamlines exam preparation by generating customized tests using user-uploaded learning materials. Integrated a retrieval-augmented generation (RAG) pipeline using ChromaDB and Together AI's LLM to create context-aware quizzes, reducing manual test prep time by over 85%. Built a robust backend with Flask and JWT-secured APIs, and a React + TypeScript frontend featuring Material UI for intuitive UX. The system supports multiple question types and intelligent evaluation, with real-time analytics identifying user strengths and gaps. Locally deployed a fully functional prototype tested with 10+ sample documents and 50+ generated questions. (*May* '25)
- **Resume Review Tool (LLMs, Mistral-7B, Hugging Face, LangChain, Streamlit)** ***(Individual)*** **Github** : Built a web-based resume analyzer that compares user resumes with job descriptions to provide ATS-style feedback and skill recommendations. Integrated the Mistral-7B-Instruct model via Hugging Face and LangChain to generate resume summaries, highlight strengths and weaknesses, and compute a match percentage with missing keyword suggestions. Designed a Streamlit interface for an intuitive user experience, enabling users to upload resumes and receive actionable, role-specific advice in under 30 seconds. (*January* '25)
- **Insurance Fraud Detection System (ML, Scikit-learn, Streamlit, Python)** ***(Individual)*** **Github** : **Developed** a fraud detection application leveraging supervised machine learning models to identify fraudulent activities. Conducted comprehensive EDA, data visualization, and feature engineering to enhance model performance. Utilized SMOTE for handling data imbalance and oversampling. Evaluated and cross-validated 8 machine learning models, selecting the best-performing model. Integrated the model with a Streamlit frontend for real-time fraud detection. Achieved the following performance metrics: Accuracy: 87.1%, Precision: 91.6%, Recall: 81.1%, F1-Score: 86.0%. Also deployed working application at this Link. (*January* '25)
- **CogniSafe (Women Safety Analytics System) (Deep Learning, PyTorch, CNN, ResNet50)** (***Team Lead***): **Developed** a real-time safety monitoring tool using an ensemble of a custom CNN and pre-trained ResNet50 for gender classification, weapon detection, and threat analysis, achieving metrics like 93.4% accuracy in gender classification and 88.5% F1-score in threat analysis. Designed an automated alert system for high-risk scenarios, enhancing public safety. Collaborated with a multidisciplinary team to integrate analytics and alert pipelines seamlessly, ensuring improved women's safety in public spaces. (*August* '24)

Activities and Leadership

- **Participated** in Smart India Hackathon 2024, leading a team in designing women's safety analytics system, *CogniSafe* , using ensemble techniques and CNN design to identify potential threats to women in isolated areas or dark areas using live footage from survilleance systems. (*September '24*)
- **Participated** in the National AgriInnovate Hackathon 2024, was a member of the team responsible for building a comprehensive and complete application for farmers to check the current market prices and trends. Integrated a prediction model that could predict the prices of crops in coming days which would help the farmers. (*August '24*)